

Measuring Like a Scientist

SI units, standard form and significant figures

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Today's plan

(slide 2)

Every lesson starts from here: what we will do, in order. Today there are four steps.

1. **Recall.** What did we measure?
2. **SI units.** One language for everyone.
3. **Standard form.** Very big, very small.
4. **Significant figures.** How many digits are true?

Then come the exercises, which you do in silence.

The case of the impossible stopwatch

(slide 3–4)

A case for Sherlock Holmes. A runner covers 100 m. A witness times him with a phone. The screen shows 11.3748 s. Holmes says: “The witness is claiming too much.”

The display is real: the app really shows four decimals. Holmes does not say that the time is false; he says that those decimals are worth nothing. Who the witness is does not matter.

Before moving on. Why does Holmes say that? Choose one. **A:** the app is broken. **B:** a finger cannot press STOP so precisely. **C:** nobody runs 100 m in 11 s. We close the case at the end.

Recall: the scale and the stopwatch

(slide 5–6)

Last lesson we looked at some instruments (figure 1). The balance reads 12.37 g.

Before moving on. How many digits are there after the decimal point? And how many in total? Then the stopwatch reading 08.46: how many digits after the point? Which shows more digits, the stopwatch or the phone in the case?

After the point there are 2 digits; in total, 4. The stopwatch reads 08.46: 2 digits after the point, that is hundredths of a second. The phone in the case shows 4 digits after the point, so the phone shows more digits. Which of the two is more reliable we do not say yet: the case closes at the end. One remark: in English texts we use the decimal point, not the comma.



Figure 1: The three displays of the recall: the balance, the stopwatch and the phone of the case.

Pressing STOP

(slide 7)

Where does the uncertainty come from? You see the runner cross the line. Your brain reacts. Your finger moves.

Before moving on. How long does all this take: about 2 s, 0.2 s or 0.002 s?

About 0.2 s. A typical visual reaction time is 0.2 s–0.25 s. This is the uncertainty in pressing the stopwatch that we met on 9 September. Keep it in mind until the end of the lesson.

Source: Jain et al., *Int. J. Appl. Basic Med. Res.* 5(2), 2015.

SI units: the base quantities

(slide 8–9)

From here on we take notes: notebook open, pen in hand. The International System of Units (SI) has seven base units; every other unit is built from them.

Quantity	Unit	Symbol
length	metre	m
mass	kilogram	kg
time	second	s
electric current	ampere	A
temperature	kelvin	K
amount of substance	mole	mol
luminous intensity	candela	cd

The first three rows (length, mass, time) are the ones we use all year.

Source: BIPM, *The International System of Units (SI Brochure)*, 9th ed., 2019.

Before moving on. The kilogram is a multiple of which unit? And which of the two, kilogram or gram, is the SI base unit of mass?

The kilogram is a multiple of the gram, but it is the kilogram that is the base unit.

Derived quantities

(slide 10)

Derived units are built from the base units with the operations.

Quantity	Unit	Built as
area	m^2	length $\times$ length
volume	m^3	length $\times$ length $\times$ length
speed	m/s	length $\div$ time
density	kg/m^3	mass $\div$ volume

Before moving on. Is density measured in kg/m^3 or in $\text{kg} \cdot \text{m}^3$?

In kg/m^3 , because density is mass *divided by* volume.

Prefixes: powers of ten

(slide 11)

Prefixes let us write very large and very small values with the same unit:

$$\begin{aligned}1 \text{ km} &= 10^3 \text{ m} = 1\,000 \text{ m} \\1 \text{ mg} &= 10^{-3} \text{ g} \\1 \mu\text{s} &= 10^{-6} \text{ s} \\1 \text{ nm} &= 10^{-9} \text{ m}\end{aligned}$$

The symbol c stands for centi and μ for micro. One practical rule: when you step down to a smaller unit, the number becomes bigger.

Before moving on. Is 5 km equal to 5 000 m or to 0.005 m?

5 000 m: the metre is a smaller unit than the kilometre, so the number is bigger.

Simple conversions

(slide 12–13)

Before moving on. To convert 2.5 km to metres, do I multiply or divide by 1000? And to convert 350 g to kilograms?

From km to m we multiply (smaller unit, bigger number); from g to kg we divide.

- 2.5 km $\rightarrow$ m: 1 km = 1 000 m, so $2.5 \times 1000 = 2\,500$ m.
- 350 g $\rightarrow$ kg: 1 kg = 1 000 g, so $350 \div 1000 = 0.350$ kg. Note that 0.350 kg keeps three significant figures, like 350 g; we come back to this in the third block.

Now a speed: 72 km/h in m/s. Convert one unit at a time.

1. $72 \text{ km/h} = \frac{72 \times 1\,000 \text{ m}}{3\,600 \text{ s}}$, because 1 km = 1 000 m and 1 h = 3 600 s.
2. $= \frac{72\,000 \text{ m}}{3\,600 \text{ s}}$: multiply first.
3. $= 20 \text{ m/s}$: then divide.

Check that the result is sensible: a car at 72 km/h does not travel at hundreds of metres per second.

Standard form

(slide 14–15)

A new block. Standard form is the way to write very large and very small numbers.

A number in standard form is $a \times 10^n$, with $1 \leq a < 10$ and n an integer.

Correct: 3.84×10^5 m. Not standard form: 38.4×10^4 m.

Before moving on. Is 38.4×10^4 in standard form?

No, because 38.4 is not between 1 and 10. The two numbers have the same value: the rule is only about the form.

Writing in standard form

(slide 16)

Two steps. First I move the point until only one non-zero digit is left in front of it; then I count the moves, and that is the exponent.

- 384 000 km: move the point 5 places left, 3.84×10^5 km.
- 0.00045 m: move the point 4 places right, 4.5×10^{-4} m.

Big number, positive power; small number, negative power. This is also the check: 0.00045 m is small, so the exponent must be negative.

Significant figures: which digits count

(slide 17–18)

The last block of content, and then the case closes. There are four rules.

1. Non-zero digits count: 12.37 g has 4.
2. Zeros between digits count: 105 s has 3.
3. Leading zeros never count: 0.0045 m has 2.
4. Final zeros after the point count: 3.00 kg has 3.

For 3.00 kg: if the balance writes two zeros, it means it guarantees them.

There is an ambiguous case: 1 300 kg. Standard form removes the ambiguity: 1.3×10^3 kg has 2 significant figures. We will meet this case in the exercises.

The calculator trap

(slide 19)

A common mistake. Example: the density of the Earth, with

Data	Value	Significant figures
mass	$6 \times 10^{24} \text{ kg}$	1
volume	$1.09 \times 10^{21} \text{ m}^3$	3

The calculator gives $5\,504.6 \text{ kg/m}^3$. The honest answer is $6 \times 10^3 \text{ kg/m}^3$.

When you multiply or divide, count the significant figures of each number in the data. Your answer gets the smallest count.

Before moving on. In the Earth example, does the answer have 5, 2 or 1 significant figures?

One: the mass has 1 and the volume has 3, and the smaller count wins. The calculator does not know how good the data are: you do.

This rule holds only for multiplication and division; sums we do not treat today. And one clarification: 11 m stays 11 m, with two significant figures. Standard form is not compulsory; it serves for very large or very small numbers, or to remove the ambiguity of final zeros, as in 1 300 kg.

Back to Holmes: case closed

(slide 20)

The answer is **B**. The reaction time is about 0.2 s, so the uncertainty is in the tenths of a second and the last three digits of 11.3748 s mean nothing. With an uncertainty of about 0.2 s, even the tenths digit is uncertain, so the most we can say is 11.4 s; 11 s would be even more honest.

Answer A is wrong: the app works, it really shows four decimals. Answer C is wrong: 11 s is a normal time for a trained sprinter.

For comparison, the world record for the 100 m, 9.58 s, is written with two decimals only, even with electronic timing.

Source: World Athletics, men's 100 m world record, Usain Bolt, Berlin 2009.

Exercises

(slide 21)

Write every step, not only the result. Give each result with the correct number of significant figures.

1. How many significant figures does each measurement have? (a) 0.0045 m (b) 105 s
(c) 3.00 kg (d) 0.02050 m (e) 4.60×10^3 m
2. Round each measurement to the number of significant figures given. (a) 8.4567 m to 3
(b) 0.003482 s to 2 (c) 1 257 kg to 2 (d) 9.96 cm to 2
3. Convert each measurement and write which physical quantity it is. (a) 2 450 mg to g
(b) 5 400 s to h (c) 90 km/h to m/s (d) 2.5 L to cm^3 (e) 1.2 g/cm^3 to kg/m^3
4. Give every answer in standard form. (a) Write 5 900 000 m. (b) Write 0.000072 s.
(c) Convert 6.4×10^3 km to m. (d) The speed of light is 3.0×10^8 m/s: convert it to km/h.
5. A pile of 500 sheets of paper is 5.0 cm thick (figure 2). Find the thickness of one sheet, in metres, in standard form.
6. The mass of the Moon is 7.3×10^{22} kg and its radius is 1 737 km (figure 3). Find its density in kg/m^3 . Volume of a sphere: $V = \frac{4}{3}\pi r^3$.

Solutions are at the end of the chapter.

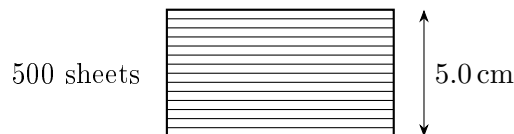


Figure 2: A pile of 500 sheets of paper, 5.0 cm thick (schematic, not to scale: only some sheets are drawn).

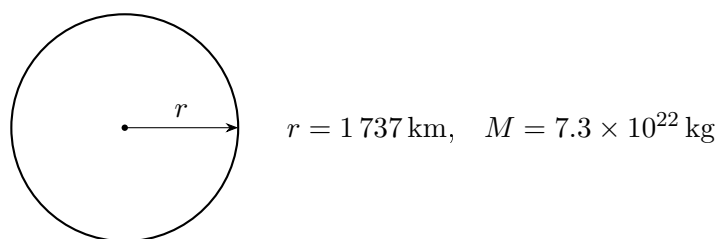


Figure 3: The Moon as a sphere of radius r (schematic).

Solutions of the exercises

(slide 21)

1. (a) 2: the leading zeros only fix the position of the point. (b) 3: the zero between two non-zero digits counts. (c) 3: the final zeros after the point count. (d) 4: starting from the 2 we have 2, 0, 5, 0, and the final zero after the point counts. (e) 3: the power of ten carries no significant figures.
2. (a) The fourth digit is 6, which is at least 5, so round up: 8.46 m. (b) The two significant figures are 3 and 4; the next digit, 8, rounds up: 0.0035 s. (c) The two significant figures are 1 and 2; the next digit, 5, rounds up to 13, and the zeros keep the value: $1\,300\text{ kg} = 1.3 \times 10^3\text{ kg}$, written in standard form to avoid ambiguity. (d) 9.96 rounds to 10, whose final zero is significant but is not visible when written 10; only $1.0 \times 10^1\text{ cm}$ makes it explicit.
3. (a) $1\text{ mg} = 10^{-3}\text{ g}$, so $2450 \times 10^{-3} = 2.450\text{ g}$: mass. (b) $1\text{ h} = 3\,600\text{ s}$, so $5400 \div 3600 = 1.5\text{ h}$: time. (c) $90 \times 1000 \div 3600 = 25\text{ m/s}$: speed. (d) $1\text{ L} = 1\text{ dm}^3 = 1\,000\text{ cm}^3$, so $2.5 \times 1000 = 2\,500\text{ cm}^3 = 2.5 \times 10^3\text{ cm}^3$: volume. (e) $1\text{ g} = 10^{-3}\text{ kg}$ and $1\text{ cm}^3 = 10^{-6}\text{ m}^3$, so $1.2 \times 10^{-3} \div 10^{-6} = 1.2 \times 10^3\text{ kg/m}^3 = 1\,200\text{ kg/m}^3$: density.
4. (a) Move the point 6 places left: $5.9 \times 10^6\text{ m}$. (b) Small number, negative exponent; move the point 5 places right: $7.2 \times 10^{-5}\text{ s}$. (c) $1\text{ km} = 10^3\text{ m}$, so $6.4 \times 10^3 \times 10^3 = 6.4 \times 10^{3+3} = 6.4 \times 10^6\text{ m}$: the exponents add. (d) $1\text{ m/s} = 3.6\text{ km/h}$, so $3.0 \times 10^8 \times 3.6 = 1.08 \times 10^9$; the datum has 2 significant figures, hence $1.1 \times 10^9\text{ km/h}$.
5. $5.0\text{ cm} = 0.050\text{ m}$; $0.050 \div 500 = 0.00010\text{ m} = 1.0 \times 10^{-4}\text{ m}$.
6. $r = 1\,737\text{ km} = 1.737 \times 10^6\text{ m}$. $r^3 = 5.241 \times 10^{18}\text{ m}^3$. $V = \frac{4}{3}\pi r^3 = 2.195 \times 10^{19}\text{ m}^3$. $\rho = m/V = 7.3 \times 10^{22}\text{ kg} \div 2.195 \times 10^{19}\text{ m}^3 = 3\,325\text{ kg/m}^3$. The mass has 2 significant figures, so $\rho = 3.3 \times 10^3\text{ kg/m}^3$.